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Case Study 13: SpicePot Restaurant Chain - Sales & Customer Analytics

Case Study 13

SpicePot Restaurant Chain - Sales & Customer Analytics

Industry: Food & Restaurant Chain

Modules: Reporting, Exploration, Dashboard, Story Making

Real-Time Business Scenario:

SpicePot is a multi-cuisine restaurant chain with 25 outlets across South India offering dine-in, takeaway, and home delivery. The operations head has observed that weekend peak hours are causing long waiting times, certain menu items are consistently underperforming, and delivery orders are increasing customer complaints due to cold food and long delivery times. IBM Cognos Analytics is being used to analyze sales, menu performance, and service quality.

Dataset / Data Available:

- Order ID, Outlet Name, Outlet City, Order Type (DineIn/Takeaway/Delivery), Order Date, Day of Week
- Menu Item, Category (Starters/Mains/Desserts/Beverages), Quantity Ordered, Item Price, Total Bill
- Order Time, Preparation Time (mins), Delivery Time (mins), Order Status (Completed/Cancelled)
- Customer ID, Customer Rating (1–5), Feedback Tag (Food Quality/Service/Delivery/Ambience), Waiter Name

SpicePot Restaurant Chain - Sales & Customer Analytics

Q1 :[Reporting] Create a Menu Performance Report showing Menu Item, Category, Total Quantity Ordered, Total Revenue, and Contribution % to Total Revenue. Apply Conditional Styling: Items contributing < 1% to revenue in Red (consider removing). Add prompts for Outlet City and Category. Include Header with SpicePot branding, Footer with page number. Use Palatino Linotype font, size 11.

SpicePot Restaurant Chain - Menu Performance Report

*

☐

Bengaluru

☒

Hyderabad

☐

Mysuru

☐

Coimbatore

☐

Chennai

☐

Kochi

Select all

Deselect all

*

☒

Beverages

☐

Desserts

☒

Mains

☐

Starters

Select all

Deselect all

SpicePot Restaurant Chain - Menu Performance Report

Menu Item	Category	Quantity Ordered	Total Bill	Contribution % to Total Revenue
Cold Coffee	Beverages	4	2649.28	17.19%
Biryani	Mains	2	124.22	0.81%
Chai	Beverages	5	1933.1	12.54%
Soft Drink	Beverages	10	1894.54	12.29%
Grilled Fish	Mains	2	476.78	3.09%
Dal Makhani	Mains	8	2509.81	16.29%
Fresh Juice	Beverages	6	2584.04	16.77%
Lassi	Beverages	5	3239.98	21.02%

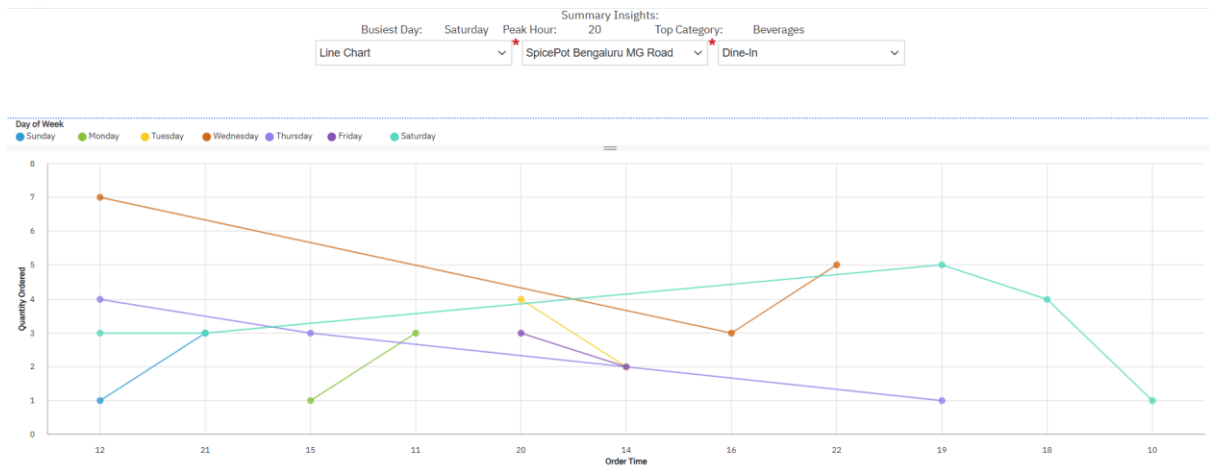
Q2 :[Reporting] Build a Delivery Performance Report listing all delivery orders with Order Time, Preparation Time, Delivery Time, Total Time, and Customer Rating. Add a calculated field: Total Delivery Time = Preparation + Delivery. Apply Conditional Styling: Total Time > 60 mins in Red, 45–60 in Amber, < 45 in Green. Summarize average delivery time by Outlet City in the footer.

SpicePot Restaurant Chain - Delivery Performance Report						
Order ID	Order Time	Preparation Time (mins)	Delivery Time (mins)	Total Delivery Time	Customer Rating (1-5)	Outlet City
ORD00042	15:04	12	40	52	4.4	Bengaluru
ORD00197	19:44	38	42	80	4.8	
ORD00116	15:45	12	31	43	1.3	
ORD00026	15:07	36	59	95	1	
ORD00029	12:39	24	39	63	5	
ORD00051	11:16	41	42	83	3.9	
ORD00185	21:14	23	57	80	4.4	Chennai
Bengaluru - Average				70.85714286		
ORD00003	18:07	20	28	48	4.7	
ORD00093	17:32	39	53	92	2.2	
ORD00133	11:10	34	25	59	1.5	
ORD00199	13:59	18	10	28	3.2	
ORD00031	12:48	45	51	96	3.1	
ORD00058	11:43	27	19	46	3.8	
ORD00074	12:28	26	38	64	2.9	
ORD00135	17:20	17	33	50	2	
ORD00146	19:13	40	29	69	3.9	Coimbatore
Chennai - Average				61.33333333		
ORD00023	18:05	15	20	35	4.9	
ORD00047	21:51	27	49	76	4.3	
ORD00138	21:01	37	36	73	4.2	
ORD00181	16:37	10	17	27	1.4	

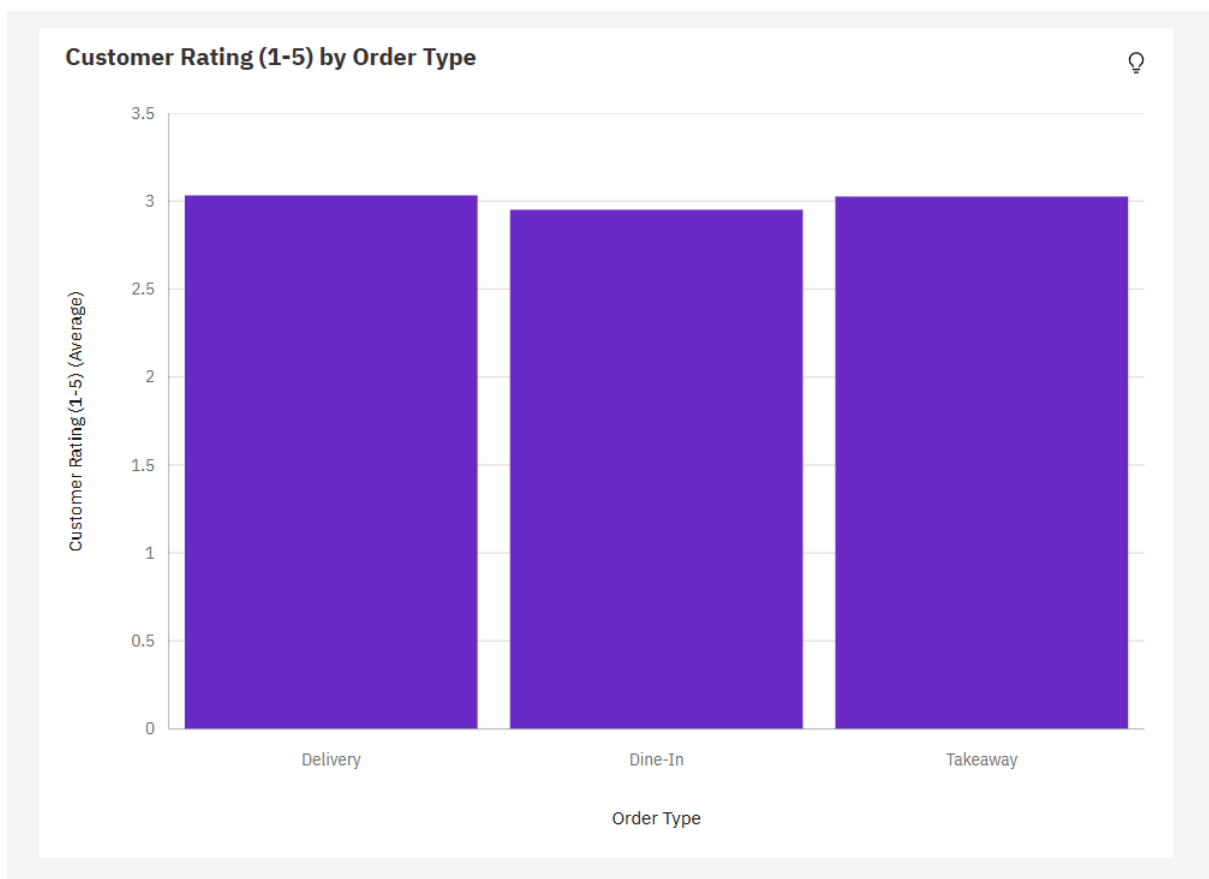
Avg Delivery Time: Bengaluru=70.86 mins | Chennai=61.33 mins | Coimbatore=57.88 mins | Hyderabad =64.64 mins | Kochi =65.12 mins | Overall - Average=63.61 mins

Q3: [Reporting] Using Report Visualize Prompt, allow Outlet Managers to select chart type (Bar or Line) to view Hourly Order Volume Trends across different days of the week. Add prompts for Outlet Name and Order Type. Summarize peak hour, busiest day, and most popular category as text objects. Apply Conditional Styling to highlight hours with Cancelled orders > 10 in Amber.

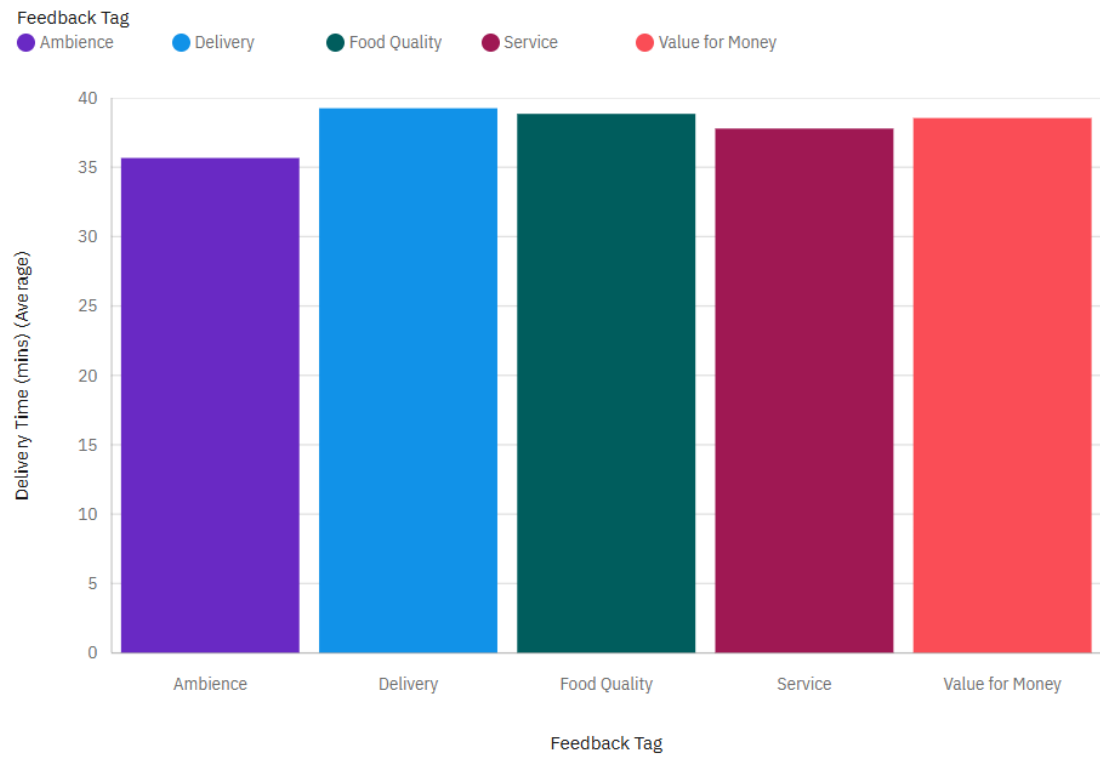




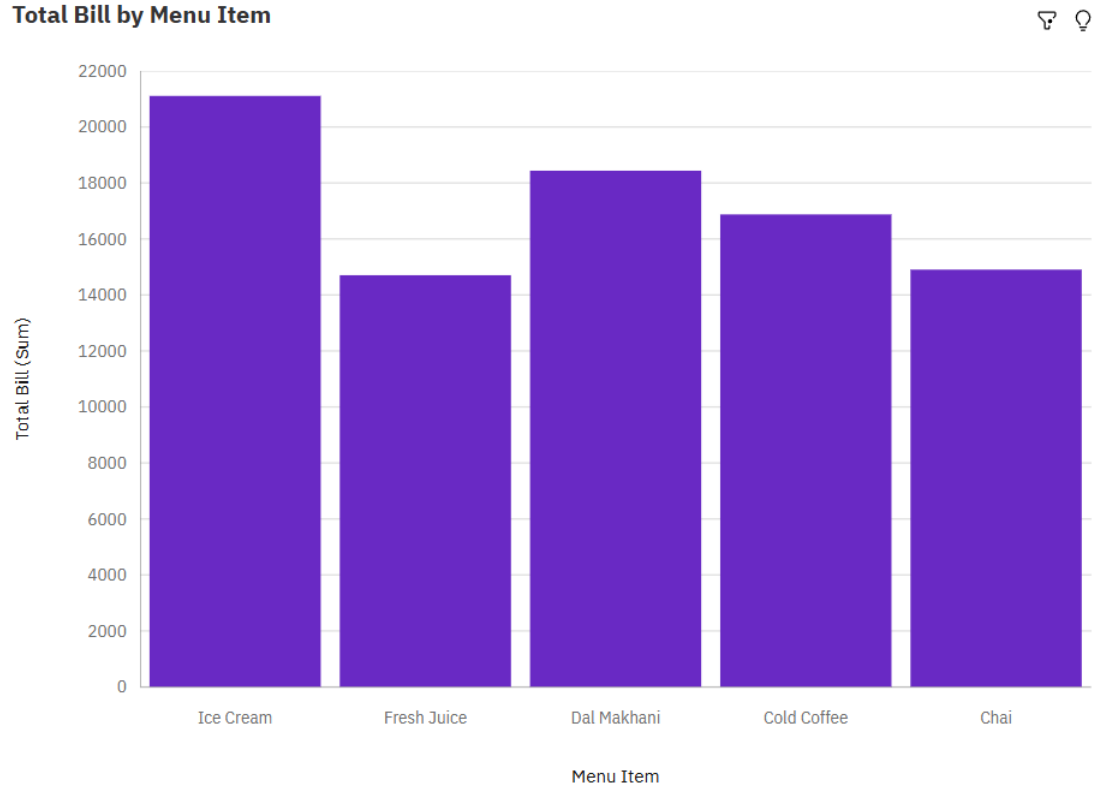
Q4: [Exploration] Use Cognos Exploration to discover patterns in the SpicePot dataset. Explore the relationship between Order Type (Dine-In/Takeaway/Delivery) and Customer Rating. Investigate which Feedback Tags appear most frequently for Delivery orders and how they correlate with Delivery Time. Identify the top 5 best-performing menu items across all outlets and document findings.



Delivery Time (mins) by Feedback Tag colored by Feedback Tag



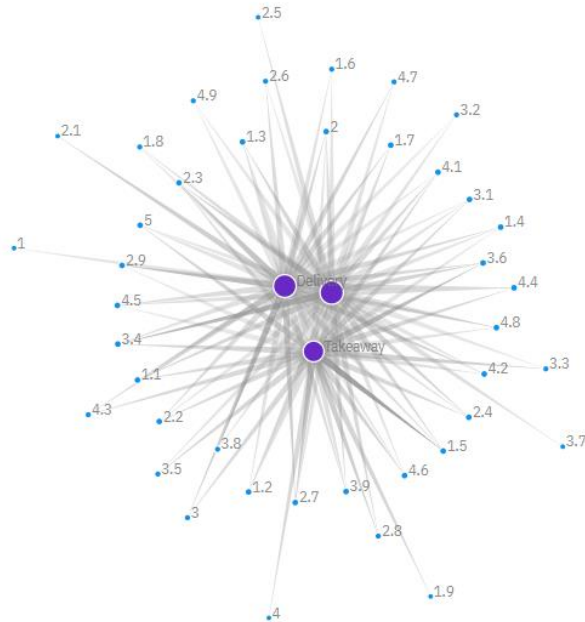
Total Bill by Menu Item



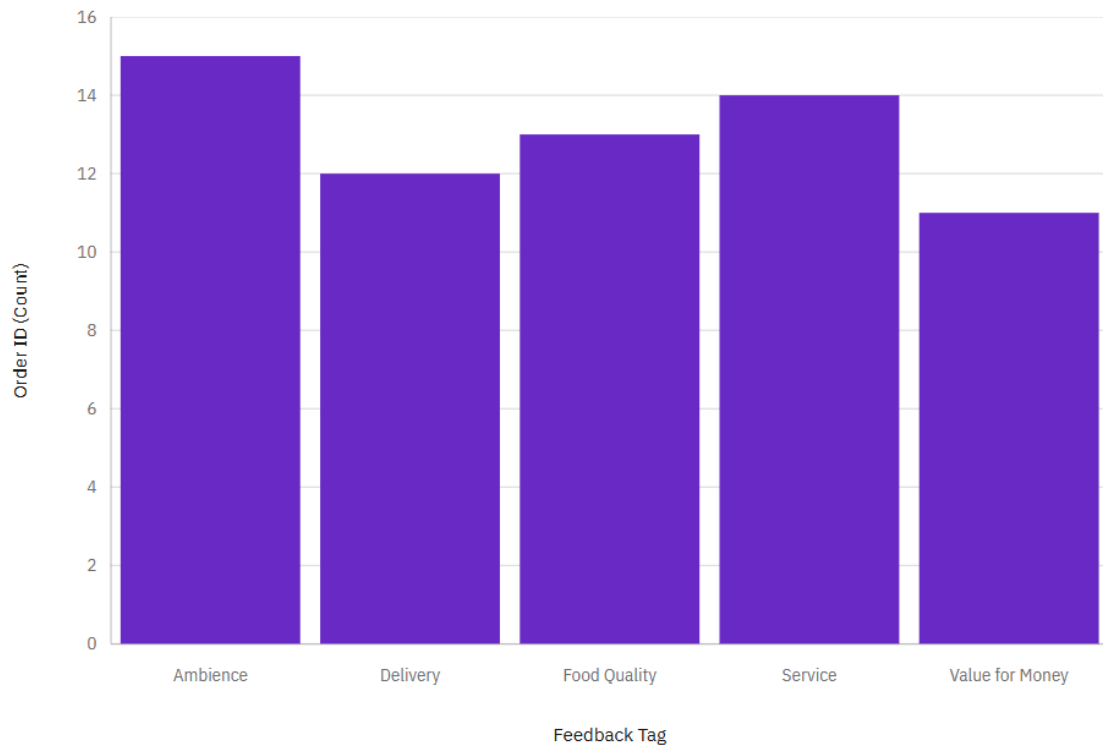
Order Type to Customer Rating (1-5) with line width Total Bill

From
● Order Type

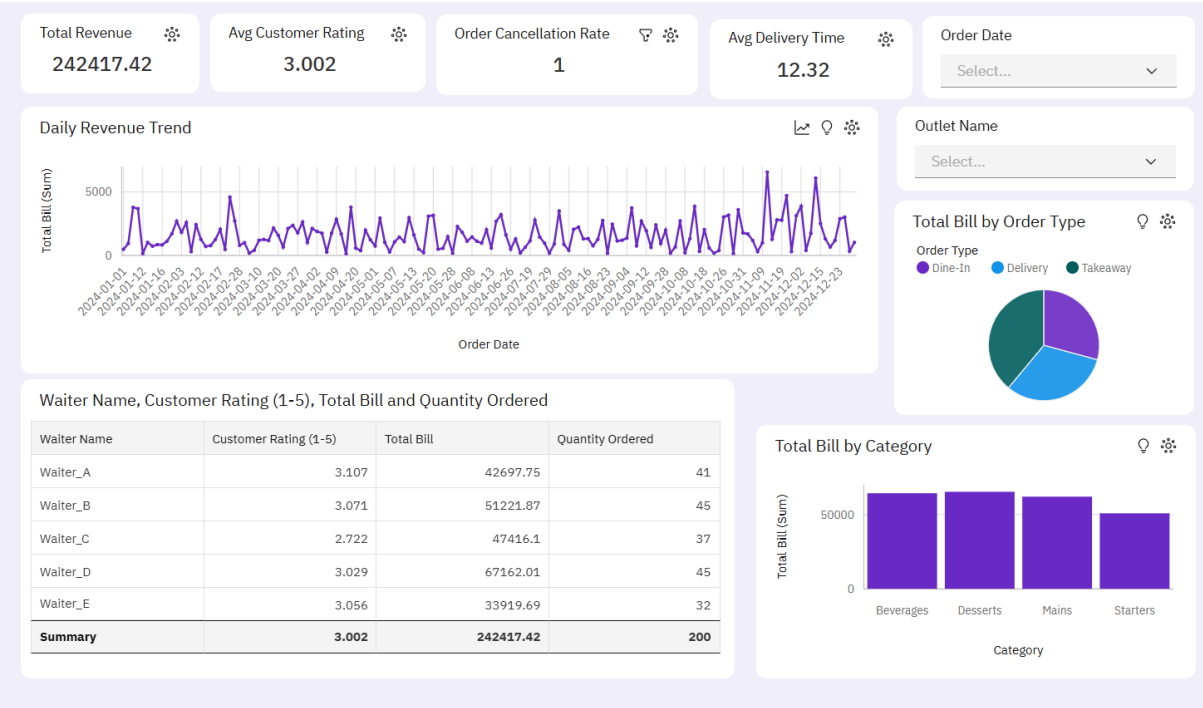
To
● Customer Rating (1-5)



Order ID by Feedback Tag



Q5: [Dashboard] Design a SpicePot Operations Dashboard with: (1) KPI tiles — Total Revenue, Average Customer Rating, Order Cancellation Rate, Average Delivery Time, (2) A Bar chart for Revenue by Menu Category, (3) A Line chart for Daily Revenue Trend, (4) A Pie chart for Order Type Distribution (Dine-In/Takeaway/Delivery), (5) A Waiter Performance table by Rating. Add Outlet and Date filters.



Total Revenue

39985.85

Avg Customer Rating

2.897

Order Cancellation Rate

1

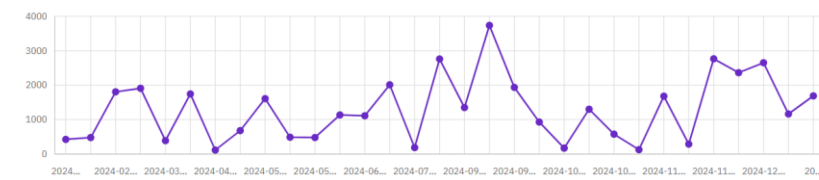
Avg Delivery Time

16.235

Order Date

Select...

Daily Revenue Trend



Outlet Name

SpicePot Hyderabad Banjara

Total Bill by Order Type

Order Type

Dine-In Delivery Takeaway



Waiter Name, Customer Rating (1-5), Total Bill and Quantity Ordered

Waiter Name	Customer Rating (1-5)	Total Bill	Quantity Ordered
Waiter_A	2.86	2642.2	5
Waiter_B	2.483	4065.32	6
Waiter_C	2.782	16624.64	11
Waiter_D	3.044	13734.62	9
Waiter_E	3.767	2919.07	3
Summary	2.897	39985.85	34

Total Bill by Category

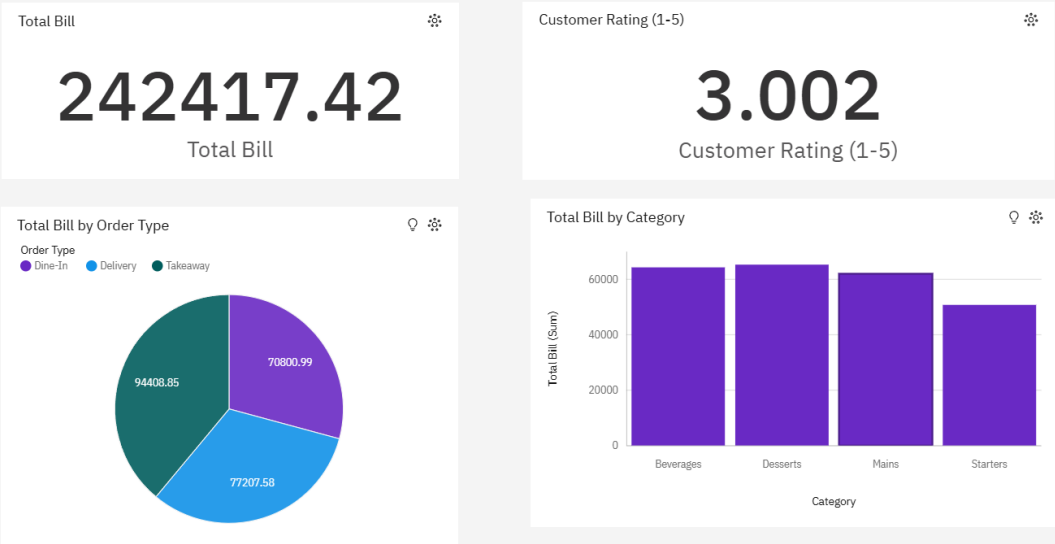


Q6: [Story Making] Create a Cognos Story for SpicePot's Monthly Operations Review:
Scene 1 — Chain-wide Sales & Order Overview, Scene 2 — Top & Bottom Performing Menu Items, Scene 3 — Delivery Performance & Complaint Analysis, Scene 4 — Customer Satisfaction by Outlet & Order Type, Scene 5 — Peak Hour & Demand Pattern Analysis, Scene 6 — Menu Optimization & Delivery Improvement Recommendations.

SpicePot Chain-wide Sales & Order Overview

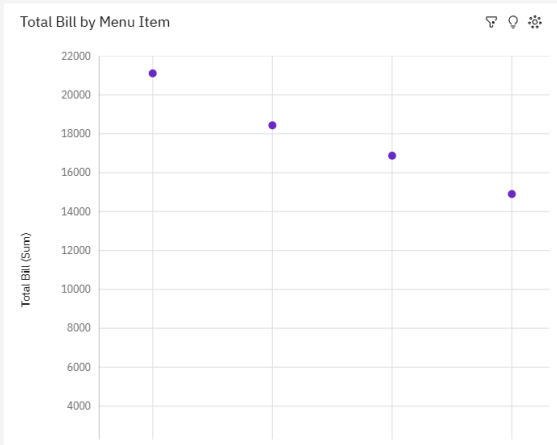
Prakash V

Monthly Operations Review — All Outlets



Top & Bottom Performing Menu Items

• Top 5 menu items by total revenue across all SpicePot



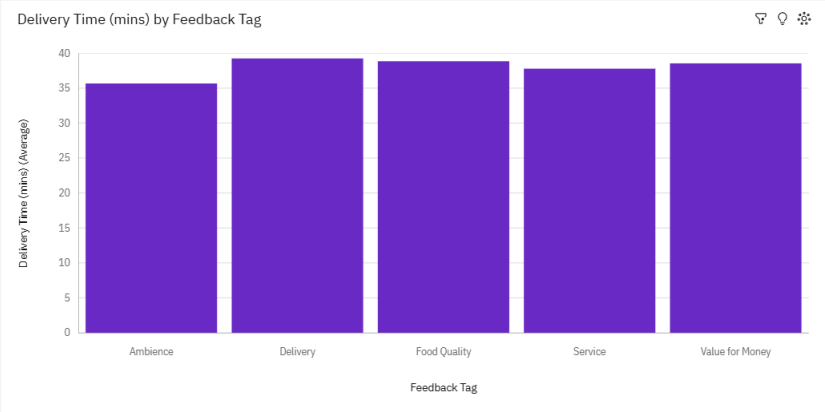
Delivery Performance & Complaint Analysis

• Delivery feedback tag shows highest avg delivery time of 39 mins. Food Quality and Service feedback also indicate dissatisfaction during delivery orders.

Delivery Time (mins)

12.32

Delivery Time (mins)



Customer Satisfaction by Outlet & Order Type

- Delivery and Takeaway orders rated higher than Dine-In. Waiter A leads with 3.107 rating.

Customer Rating (1-5)

3.002

Customer Rating (1-5)

Customer Rating (1-5) by Order Type



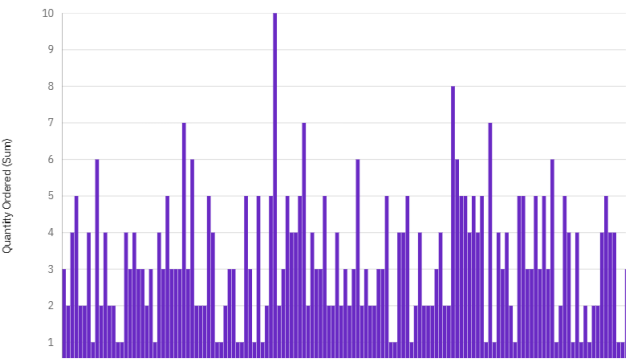
Waiter Name and Customer Rating (1-5)

Waiter Name	Customer Rating (1-5)
Waiter_A	3.107
Waiter_B	3.071
Waiter_C	2.722
Waiter_D	3.029
Waiter_E	3.056
Summary	3.002

Peak Hour & Demand Pattern Analysis

- Saturday records highest orders at 125. Peak ordering periods identified across weekdays and weekends. Wednesday and Thursday show strong mid-week demand.

Quantity Ordered by Order Time



Quantity Ordered by Day of Week



Menu Optimization & Delivery Improvement Recommendations

Menu Optimization:

- Promote: Ice Cream, Dal Makhani, Cold Coffee
- Review: Gobi Manchurian, Soft Drink
- Expand Desserts category
- Introduce combo offers

Delivery Improvement:

- Target delivery under 30 mins
- Address negative feedback tags
- Improve food quality during transit
- Add delivery tracking system

Total Bill

242417.42

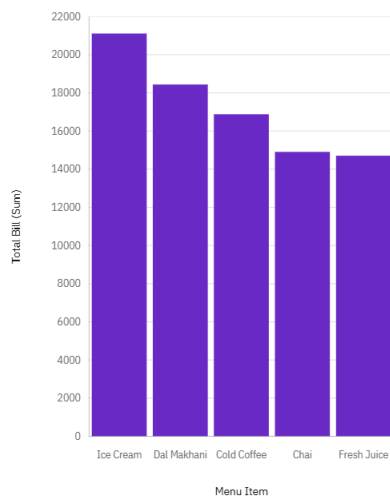
Total Bill

Customer Rating (1-5)

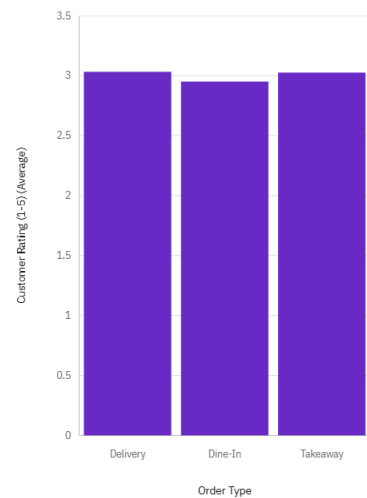
3.002

Customer Rating (1-5)

Total Bill by Menu Item



Customer Rating (1-5) by Order Type



SpicePot Web application

Program:

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
<meta charset="UTF-8">
```

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
<title>SpicePot Intelligence Hub</title>
```

```
<link
```

```
href="https://fonts.googleapis.com/css2?family=Playfair+Display:wght@700;900&family=DM+Sans:wght@400;500;700&family=JetBrains+Mono&display=swap" rel="stylesheet">
```

```
<style>
```

```
:root { --saffron: #F4841F; --ember: #E84E10; --gold: #E8B84B; --cream: #FDF6EC; --charcoal: #1A1612; --border: rgba(244,132,31,0.2); --glass: rgba(45,31,15,0.7); }
```

```
* { margin: 0; padding: 0; box-sizing: border-box; font-family: 'DM Sans', sans-serif; }
```

```
body { background: var(--charcoal); color: var(--cream); overflow-x: hidden; }
```

```
.flex { display: flex; align-items: center; }
```

```
.grid { display: grid; gap: 20px; }
```

```

.hidden { display: none !important; }

.mono { font-family: 'JetBrains Mono', monospace; font-size: 10px; letter-spacing: 2px;
text-transform: uppercase; }

/* Layout */

#login-screen, #app { min-height: 100vh; }

#login-screen { display: flex; justify-content: center; align-items: center; background:
radial-gradient(circle at 80% 20%, #2d1f0f, #1a1612); }

.card { background: var(--glass); border: 1px solid var(--border); border-radius: 4px;
padding: 30px; backdrop-filter: blur(15px); }

header { justify-content: space-between; padding: 15px 40px; background:
rgba(26,22,18,0.9); border-bottom: 1px solid var(--border); sticky; top: 0; z-index: 10; }

nav { background: rgba(26,22,18,0.5); border-bottom: 1px solid var(--border); padding:
0 40px; }

.tab-btn { padding: 18px 25px; background: none; border: none; border-bottom: 2px
solid transparent; color: #8c7b6b; cursor: pointer; font-weight: 500; transition: 0.3s; }

.tab-btn.active { color: var(--saffron); border-bottom-color: var(--saffron); }

/* Dashboard & Tables */

.kpi-grid { grid-template-columns: repeat(auto-fit, minmax(240px, 1fr)); padding: 40px;
}

.kpi-card { padding: 25px; border-left: 3px solid var(--saffron); }

.kpi-val { font-family: 'Playfair Display'; font-size: 36px; margin: 10px 0; background:
linear-gradient(to right, #fff, var(--gold)); -webkit-background-clip: text; -webkit-text-fill-
color: transparent; }

.main-grid { grid-template-columns: 1.5fr 1fr; padding: 0 40px 40px; }

.data-table { width: 100%; border-collapse: collapse; margin-top: 15px; font-size: 13px;
}

.data-table th { text-align: left; padding: 12px; color: var(--saffron); border-bottom: 1px
solid var(--border); font-weight: 400; }

.data-table td { padding: 12px; border-bottom: 1px solid rgba(255,255,255,0.05); }

```

```

/* Components */

.input { width: 100%; padding: 12px; background: rgba(0,0,0,0.3); border: 1px solid
var(--border); color: white; margin: 10px 0; outline: none; }

.btn { width: 100%; padding: 14px; background: linear-gradient(135deg, var(--saffron),
var(--ember)); border: none; color: white; cursor: pointer; font-weight: 700; border-radius:
2px; }

.status { padding: 3px 8px; border-radius: 12px; font-size: 10px; font-weight: 700; }

.completed { background: rgba(74,222,128,0.1); color: #4ade80; }

.cancelled { background: rgba(248,113,113,0.1); color: #f87171; }

iframe { width: 100%; height: 800px; border: none; border-radius: 4px; background:
#fff; }

</style>
</head>
<body>

<div id="login-screen">
  <div class="card" style="width: 420px; text-align: center;">
    <span style="font-size: 40px;">🔒</span>
    <h2 style="font-family: 'Playfair Display'; font-size: 32px; margin: 10px 0;">Analyst
Login</h2>
    <p style="color: #8c7b6b; margin-bottom: 25px;">SpicePot Intelligence Hub v2.0</p>
    <input type="text" id="user" class="input" placeholder="User ID (admin)">
    <input type="password" id="pass" class="input" placeholder="Password (admin123)">
    <button class="btn" onclick="auth()">ACCESS DASHBOARD →</button>
    <p id="err" class="hidden" style="color: #f87171; margin-top: 15px; font-size:
12px;">Invalid Credentials</p>
  </div>
</div>

<div id="app" class="hidden">
  <header class="flex">

```

```
<div class="flex" style="gap:15px">

  <div style="background:var(--saffron); padding:5px; border-radius:5px">🍵</div>

  <div><b style="font-size:20px; font-family:'Playfair Display'">SpicePot</b><span
class="mono" style="color:var(--saffron); margin-left:10px">Operational
Intelligence</span></div>

</div>

<button class="mono" style="background:none; border:1px solid var(--border);
color:var(--gold); padding:5px 15px; cursor:pointer"
onclick="location.reload()">Logout</button>

</header>

<nav class="flex">

  <button class="tab-btn active" onclick="tab('overview')">Operations
Overview</button>

  <button class="tab-btn" onclick="tab('explorer')">Raw Data Explorer</button>

  <button class="tab-btn" onclick="tab('cognos')">IBM Cognos Embed</button>

</nav>

<div id="overview" class="content">

  <div class="kpi-grid grid">

    <div class="card kpi-card"><p class="mono">Total Orders</p><div class="kpi-
val">6,400</div><small style="color:#4ade80">▲ 12.4% vs LY</small></div>

    <div class="card kpi-card"><p class="mono">Revenue</p><div class="kpi-
val">₹28.4L</div><small style="color:#4ade80">▲ 8.7% Growth</small></div>

    <div class="card kpi-card"><p class="mono">Avg Rating</p><div class="kpi-
val">3.6</div><small style="color:#f87171">▼ 0.2 points</small></div>

    <div class="card kpi-card"><p class="mono">Order Health</p><div class="kpi-
val">74%</div><small style="color:#4ade80">▲ 3.1% Success</small></div>

  </div>

  <div class="main-grid grid">

    <div class="card">

      <h3 class="mono" style="color:var(--gold)">Live Transactions</h3>

      <table class="data-table" id="table-mini"></table>

    </div>

  </div>

</div>
```

</div>

<div class="card">

<h3 class="mono" style="color:var(--gold)">City Performance</h3>

<div id="bar-chart" style="margin-top:25px"></div>

</div>

</div>

</div>

<div id="explorer" class="content hidden" style="padding:40px">

<div class="card">

<div class="flex" style="justify-content:space-between; margin-bottom:20px">

<h3 class="mono">Dataset Search</h3>

<input type="text" class="input" style="width:300px; margin:0"
placeholder="Search item, city, or ID..." oninput="search(this.value)">

</div>

<div style="max-height:600px; overflow-y:auto">

<table class="data-table" id="table-full"></table>

</div>

</div>

</div>

<div id="cognos" class="content hidden" style="padding:40px">

<div class="card" style="padding:10px">

<div class="flex" style="padding:10px; justify-content:space-between">

Live Server: ap2.ca.analytics.ibm.com

● Connected

</div>

<iframe

src="https://ap2.ca.analytics.ibm.com/bi/?perspective=dashboard&pathRef=.my_folders%2F5th%2Bquestion%2Bdashboard&closeWindowOnLastView=true&ui_appbar=false&ui_navbar=false&shareMode=embedded&action=view"></iframe>

</div>

</div>

</div>

<script>

// Logic: Data Generation

const cities = ['Chennai', 'Bengaluru', 'Hyderabad', 'Kochi', 'Coimbatore', 'Mysuru'];

const items = ['Biryani', 'Butter Chicken', 'Fish Fry', 'Paneer Tikka', 'Gulab Jamun', 'Chai'];

const data = Array.from({length: 100}, (_, i) => ({

id: `ORD\${202400 + i}`,

city: cities[i % cities.length],

item: items[i % items.length],

bill: (Math.random() * 1500 + 200).toFixed(2),

status: Math.random() > 0.15 ? 'Completed' : 'Cancelled'

}));

function auth() {

const p = document.getElementById('pass').value;

if(p === 'admin123') {

document.getElementById('login-screen').classList.add('hidden');

document.getElementById('app').classList.remove('hidden');

init();

} else { document.getElementById('err').classList.remove('hidden'); }

}

function tab(id) {

document.querySelectorAll('.content').forEach(c => c.classList.add('hidden'));

document.getElementById(id).classList.remove('hidden');

document.querySelectorAll('.tab-btn').forEach(b => b.classList.remove('active'));

event.currentTarget.classList.add('active');

}

```

function draw(list = data) {

  const head = `<tr><th class="mono">Order ID</th><th class="mono">Region</th><th
class="mono">Menu Item</th><th class="mono">Bill</th><th
class="mono">Status</th></tr>`;

  const rows = d => d.map(r => `<tr><td class="mono" style="color:var(--
gold)">${r.id}</td><td>${r.city}</td><td>${r.item}</td><td>₹${r.bill}</td><td><span
class="status ${r.status.toLowerCase()}>${r.status}</span></td></tr>`).join("");

  document.getElementById('table-mini').innerHTML = head + rows(list.slice(0, 10));

  document.getElementById('table-full').innerHTML = head + rows(list);


  document.getElementById('bar-chart').innerHTML = cities.map(c => {

    const val = Math.random() * 80 + 20;

    return `<div style="margin-bottom:18px"><div class="flex" style="justify-
content:space-between"><small
class="mono">${c}</small><small>${val.toFixed(0)}%</small></div>

    <div style="height:6px; background:rgba(255,255,255,0.05); margin-top:5px; border-
radius:10px"><div style="height:100%; width:${val}%; background:var(--saffron); border-
radius:10px; box-shadow:0 0 10px var(--saffron)"></div></div></div>`

    }).join("");

  }

  function search(v) {

    const filtered = data.filter(r => r.city.toLowerCase().includes(v.toLowerCase()) ||
r.item.toLowerCase().includes(v.toLowerCase()) || r.id.includes(v));

    draw(filtered);

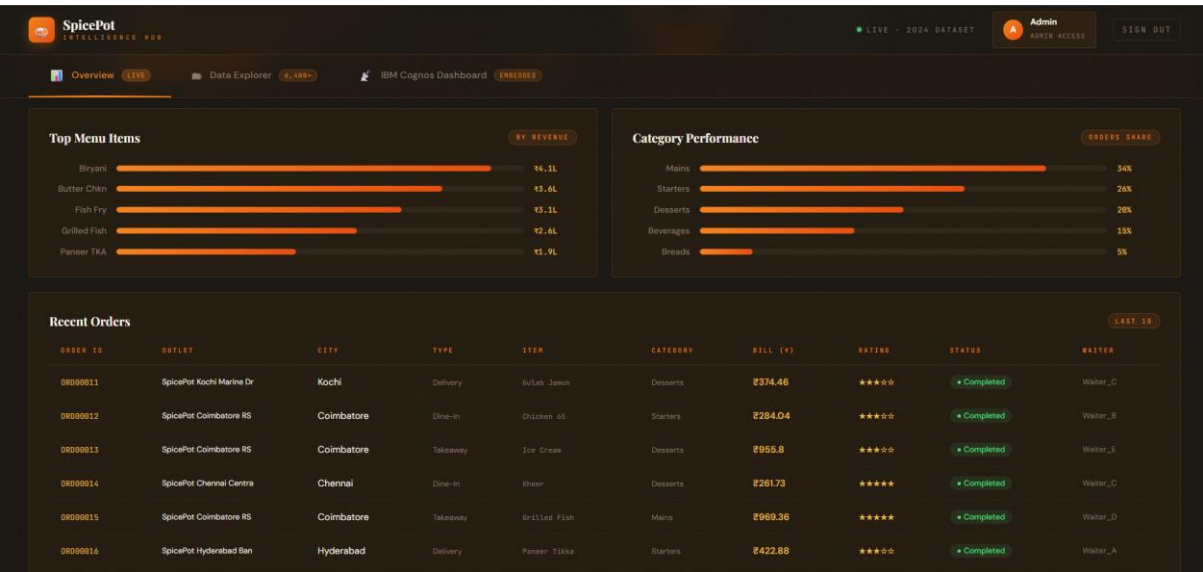
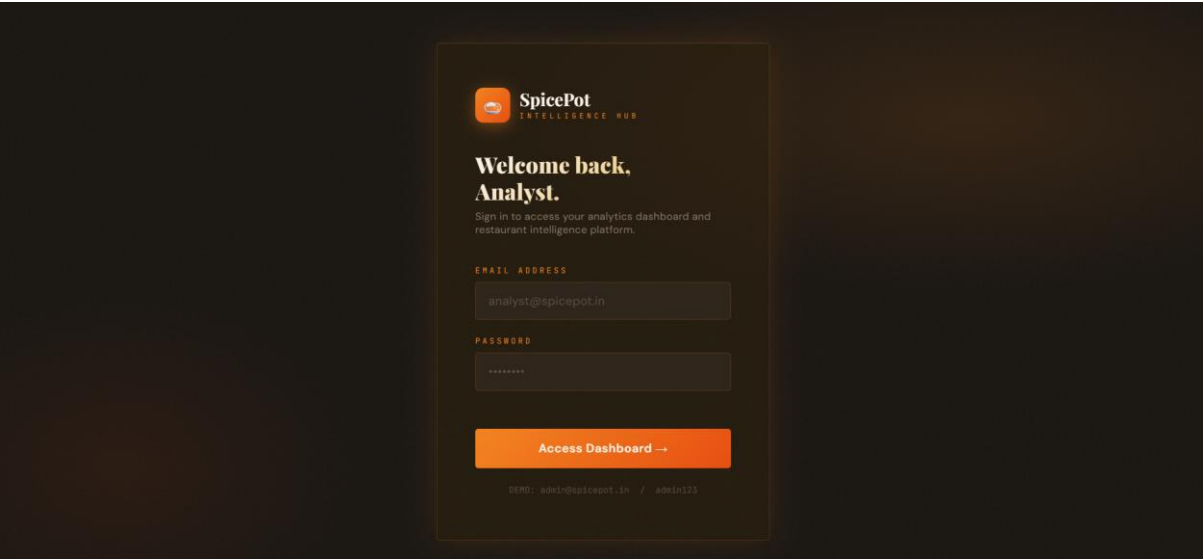
  }

  function init() { draw(); }

</script>
</body>
</html>

```

OUTPUT:



SpicePot

LIVE2024 DATASETAdminADMIN ACCESSSIGN OUT

OverviewLIVE

Data Explorer6,400+

IBM Cognos DashboardEMBEDDED

RAW DATA ACCESS

Data Explorer

Browse, filter, and search across all 6,400+ orders from the SpicePot dataset.

Search by ID, item, city...

All Cities

All Order Types

All Statuses

Export CSV

Showing 1-12 of 200 records

Dataset: SpicePot 2024 - IBM Cognos

| ORDER ID | OUTLET | CITY | TYPE | DATE | MENU ITEM | CATEGORY | QTY | TOTAL BILL | ORDER TIME | STATUS | RATING | WAITER |
|----------|--------------------------|------------|----------|------------|----------------|----------|-----|------------|------------|-----------|--------|----------|
| 00000011 | SpicePot Kochi Marine Dr | Kochi | Delivery | 2024-10-24 | Gulab Jamun | Desserts | 2 | \$374.46 | 19:14 | Completed | 3.5 | Waiter_C |
| 00000012 | SpicePot Coimbatore RS | Coimbatore | Dine-In | 2024-11-26 | Chicken 65 | Starters | 4 | \$284.04 | 20:32 | Completed | 3.3 | Waiter_B |
| 00000013 | SpicePot Coimbatore RS | Coimbatore | Takeaway | 2024-07-03 | Ice Cream | Desserts | 3 | \$955.8 | 14:18 | Completed | 3.3 | Waiter_E |
| 00000014 | SpicePot Chennai Centre | Chennai | Dine-In | 2024-06-05 | Kheer | Desserts | 1 | \$261.73 | 19:34 | Completed | 4.7 | Waiter_C |
| 00000015 | SpicePot Coimbatore RS | Coimbatore | Takeaway | 2024-10-06 | Grilled Fish | Mains | 3 | \$969.36 | 13:50 | Completed | 4.6 | Waiter_D |
| 00000016 | SpicePot Hyderabad Ran | Hyderabad | Delivery | 2024-01-11 | Paneer Tikka | Starters | 3 | \$422.88 | 22:22 | Completed | 3.4 | Waiter_A |
| 00000017 | SpicePot Coimbatore RS | Coimbatore | Dine-In | 2024-08-10 | Butter Chicken | Mains | 3 | \$2,045.43 | 12:25 | Completed | 2.4 | Waiter_D |
| 00000018 | SpicePot Chennai Centre | Chennai | Dine-In | 2024-05-11 | Fish Fry | Starters | 2 | \$1,260.78 | 13:47 | Completed | 2 | Waiter_E |
| 00000019 | SpicePot Mysuru Vijayan | Mysuru | Takeaway | 2024-09-28 | Ice Cream | Desserts | 5 | \$1,974.75 | 16:05 | Completed | 3.4 | Waiter_E |

SpicePot

LIVE2024 DATASETAdminADMIN ACCESSSIGN OUT

OverviewLIVE

Data Explorer6,400+

IBM Cognos DashboardEMBEDDED

IBM Cognos Dashboard

Embedded live dashboard powered by IBM Cognos Analytics - interactive, filterable, and real-time.

RefreshFullscreenOpen in IBM Cognos

https://api.cx.ibmcloud.com/api/v1/spicepot-analytics-ibm-cognos/ibm-cognos-dashboard

LIVE

100%AnalyticsFilters

Tab 1

Total Revenue242417.42Total Bill

Avg Customer Rating3.002Customer Rating (1-5)

Order Cancellation Rate1Order Status

Avg Delivery Time12.32Delivery Time (mins)

Order DateSelect...

Daily Revenue Trend

Outlet NameSelect...

Total Bill by Order Type

Order TypeDine-InDeliveryTakeaway

94400.8570800.99

SpicePot

LIVE2024 DATASETAdminADMIN ACCESSSIGN OUT

OverviewLIVE

Data Explorer6,400+

IBM Cognos DashboardEMBEDDED

Tab 1

Daily Revenue Trend

Outlet NameSelect...

Total Bill by Order Type

Order TypeDine-InDeliveryTakeaway

70800.9977207.58

Waiter Name, Customer Rating (1-5), Total Bill and Quantity Ordered

| Waiter Name | Customer Rating (1-5) | Total Bill | Quantity Ordered |
|-------------|-----------------------|------------|------------------|
| Waiter_A | 3.107 | 42697.75 | 41 |
| Waiter_B | 3.071 | 51221.87 | 45 |
| Waiter_C | 2.722 | 47416.1 | 37 |
| Waiter_D | 3.029 | 67162.01 | 45 |
| Waiter_E | 3.056 | 33919.69 | 32 |
| Summary | 3.002 | 242417.42 | 200 |

Total Bill by Category

CategoryBeveragesDessertsMainsStarters